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Title

Semantic Model

Sources

Transformations

The Economy

Capital Rotation

Crisis Funding

Shifting Ground

Playbook

From Raw Data to Executive Insight

This report connects 11 data sources, 100K+ records, and 50 years of economic history into a single semantic model. Built on Federal Reserve FRED data, Crunchbase venture records, and US Treasury figures — it transforms complex financial signals into a narrative any stakeholder can follow.



Semantic Model



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This report is built on a Microsoft Fabric warehouse with a full three-layer data architecture.

1) Raw data is ingested from external sources including FRED, Crunchbase, and curated CSVs into a staging area that holds the data exactly as received.

2) A transformation layer joins and cleans the staged data — forward-filling sparse series, aligning date grains, and resolving economic event overlaps.

3) A data mart layer sits on top of that, pre-calculating year-over-year changes, crisis flags, and sector mappings so Power BI reads clean, ready-to-use data without performing joins at query time. The semantic model connects directly to the mart layer only.

Semantic Model

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Transformations

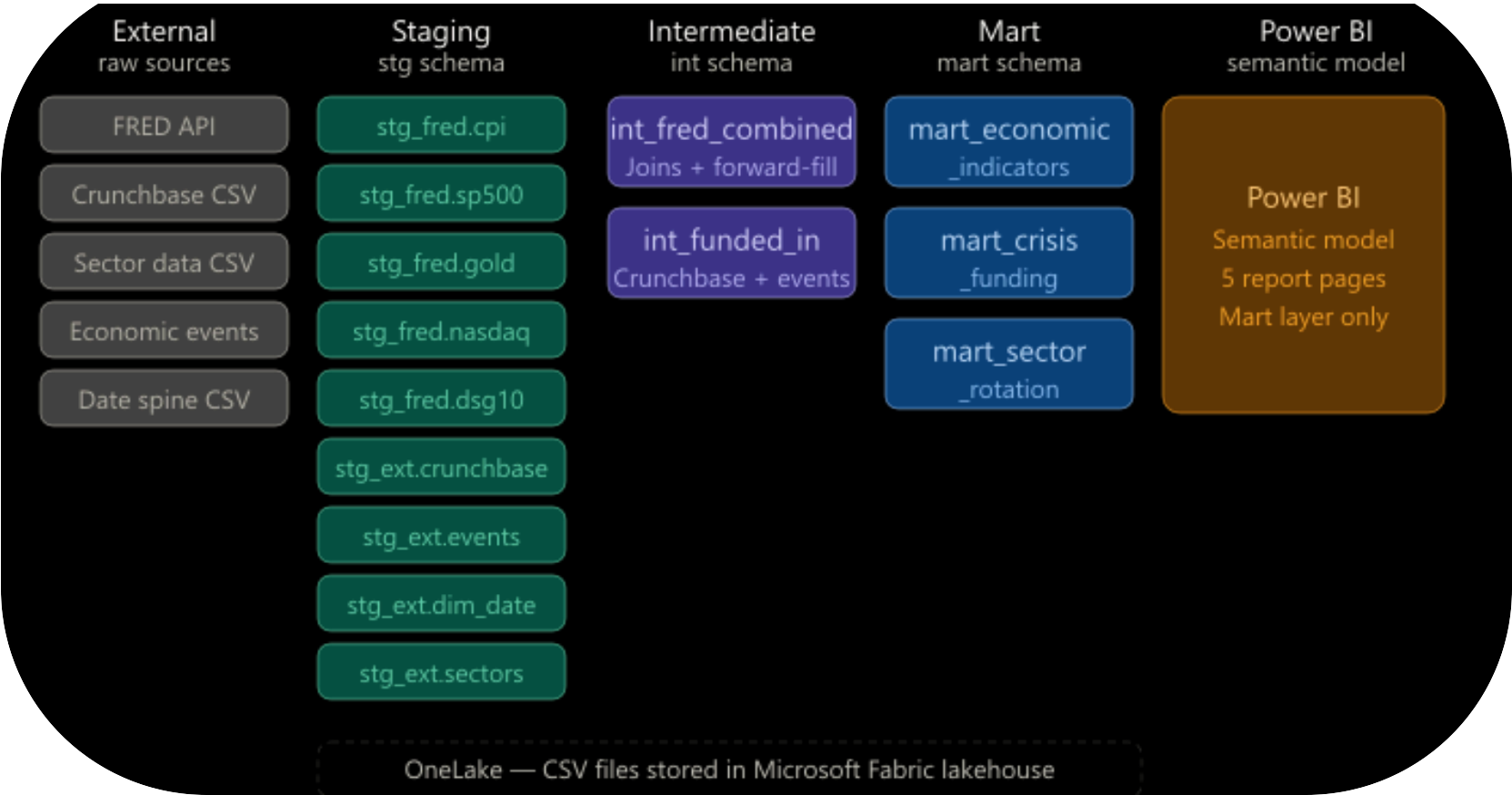
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Transformations



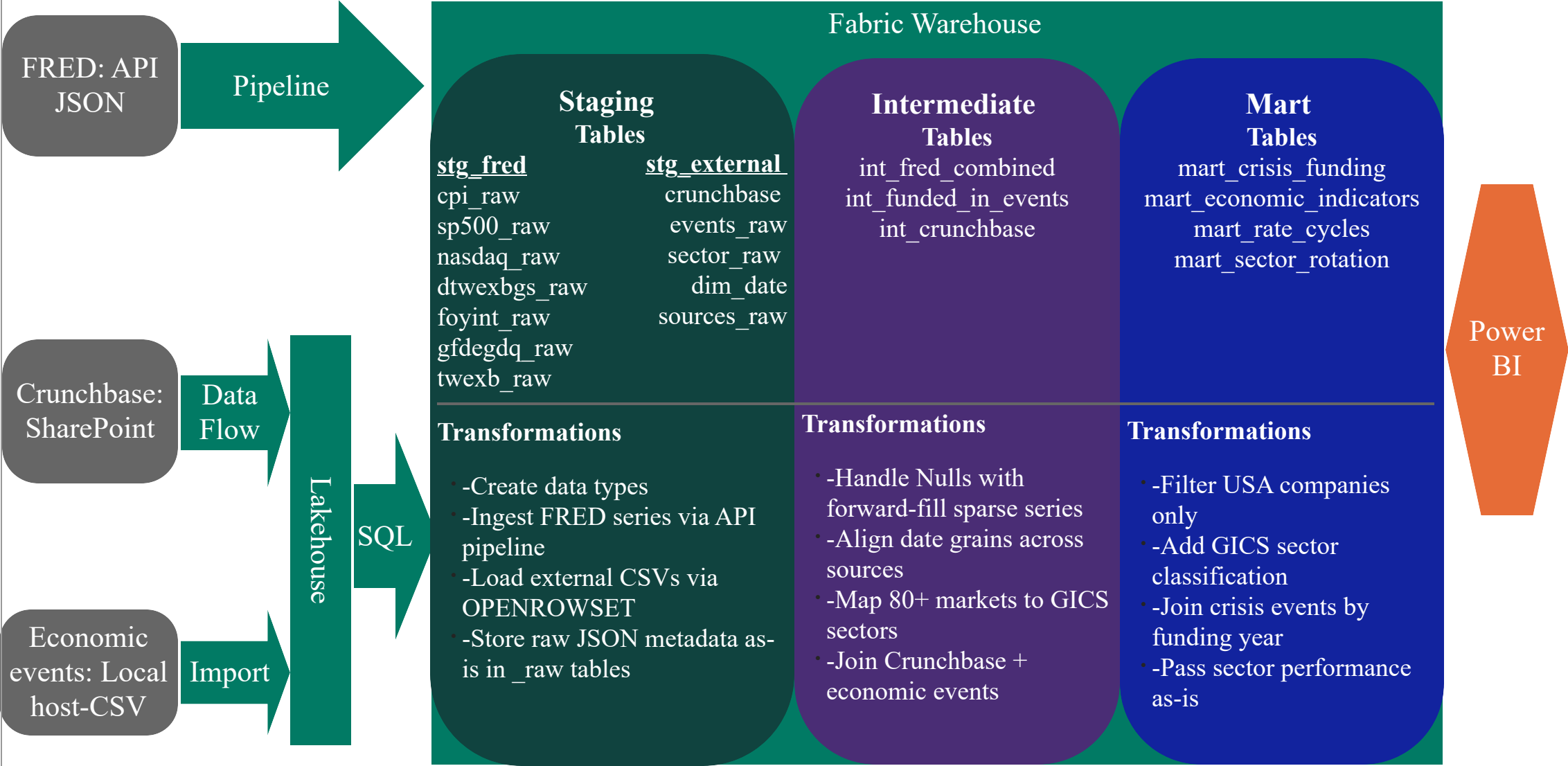
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Built on Microsoft Fabric using three distinct ingestion patterns: a Data Factory pipeline pulling live economic data from the FRED API, a Dataflow Gen2 connector sourcing startup funding data from SharePoint, and OPENROWSET queries loading curated CSVs directly from OneLake. All data lands in a three-layer warehouse architecture — staging, intermediate, and mart — before being served to a Power BI semantic model.

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50 Years of Rate Cycles — The Economy Always Comes Back



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The Federal Reserve has raised and lowered interest rates through every crisis of the last 50 years. Each time, markets fell, gold rose, and headlines declared the end of prosperity. Each time, they were wrong. The investors who understood the cycle and stayed invested captured every recovery. This chart traces that pattern — not to predict the future, but to show that the pattern has never failed to reset.

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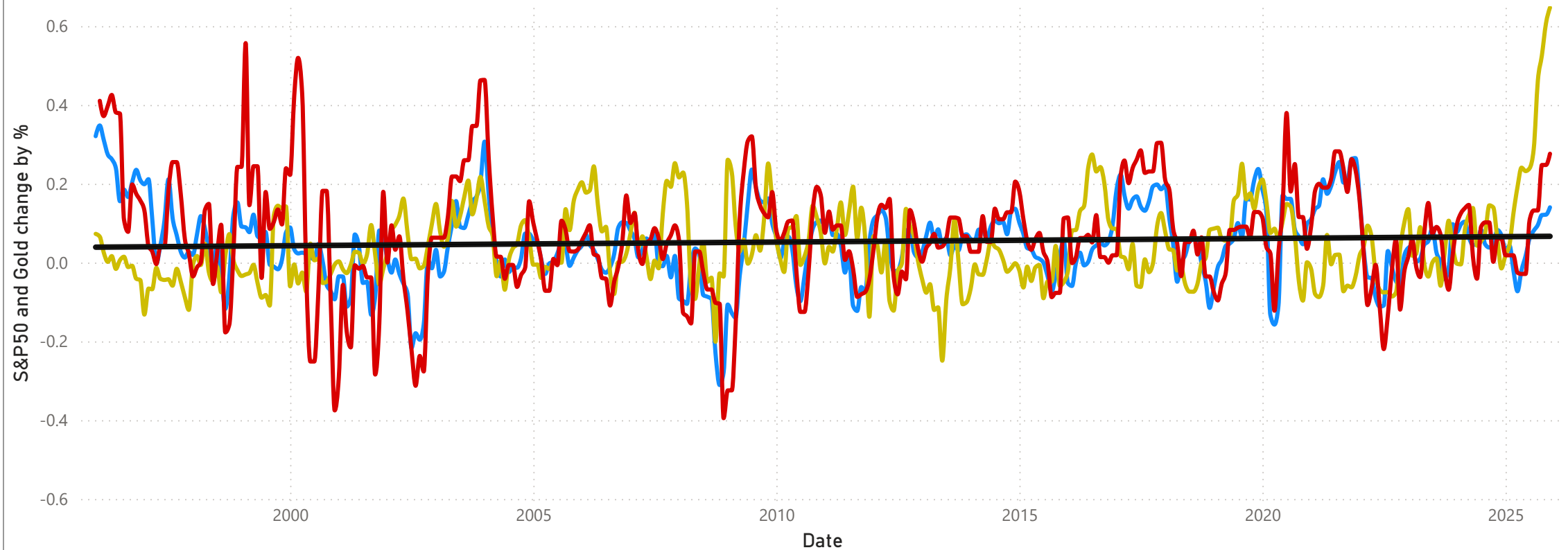
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50 Year Rate Cycle

● SP500 ● Gold ● NASDAQ





Recessions Don't Stop Capital — They Concentrate It



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When the economy contracts, conventional wisdom says freeze. The data says rotate. Venture funding doesn't disappear during downturns — it concentrates into the sectors investors trust most. The pie gets smaller but the slice going to core sectors gets bigger. Every crisis in this dataset created the conditions for the next decade's winners. The question was never whether to invest — it was where.

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Economic contractions don't eliminate investment — they filter it. When credit tightens and risk tolerance drops, capital rotates away from speculative and discretionary sectors toward those with proven demand, recurring revenue, and structural necessity. Academic research supports this pattern. Sarwar, Mateus, and Todorovic (2017) demonstrated using Fama-French five-factor models that non-cyclical sectors generate statistically significant positive alpha during monetary tightening cycles while cyclical sectors underperform. Fidelity's Asset Allocation Research Team, analyzing 12 business cycles since 1945, found that Health care, Consumer Staples, and Utilities consistently outperform during recession phases while Technology, Financials, and Consumer Discretionary decline.

The mechanism is consistent across cycles: sophisticated investors rotate, they don't retreat. Understanding where capital consolidates during stress is the foundation of the framework on the next page.

Events		Sector Performance During Economic Downturns				
COVID Crash	Financial Crisis	Event	Year	Rank	Return %	GICS Sector
		COVID Crash	2020	Top	44	Technology
		COVID Crash	2020	Second Top	14	Health Care
		COVID Crash	2020	Second Bottom	-4	Financials
		COVID Crash	2020	Bottom	-37	Energy
	Rate Hike Cycle	Dot-com Crash				

Sarwar G., Mateus C., Todorovic N. (2017). US sector rotation with five-factor Fama-French alphas. Journal of Asset Management. Fidelity Investments AART Business Cycle Framework (2021)



Companies Funded in Crisis Economies by Sector



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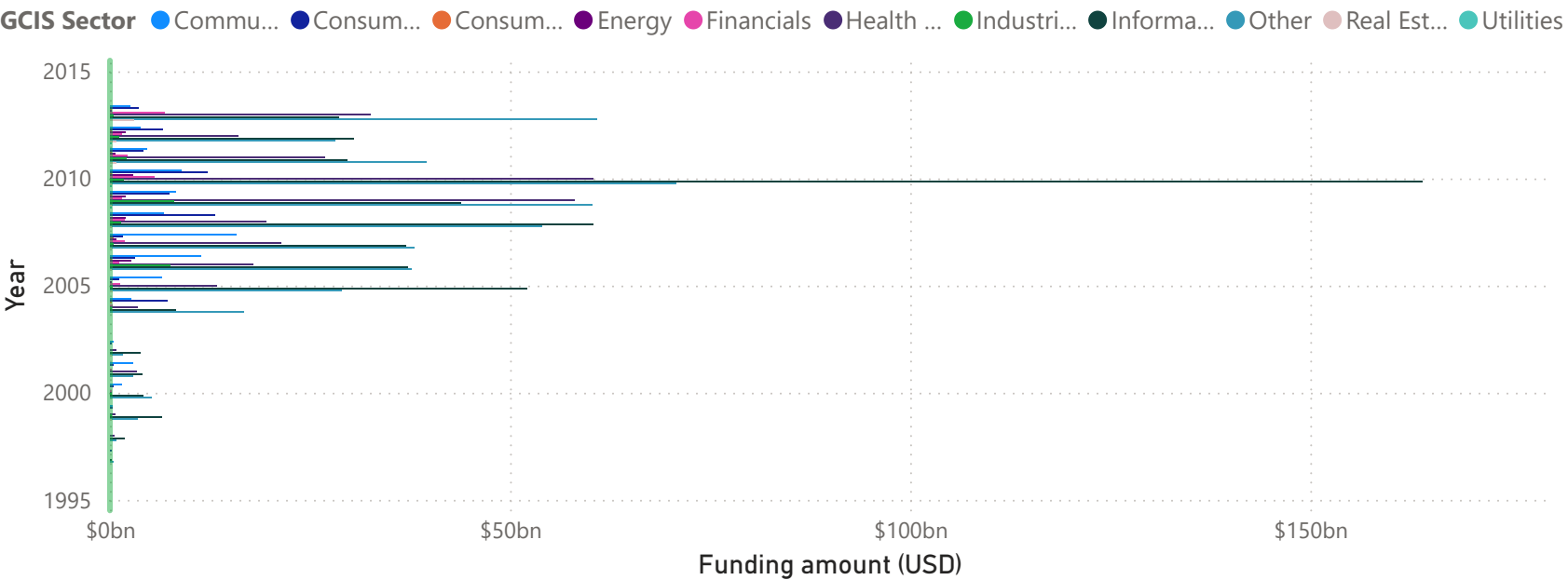
Fear doesn't stop money — it redirects it. At the peak of every crisis in this dataset, capital kept flowing into companies that would define the next era. The dot-com crash funded the infrastructure of the modern internet. The financial crisis funded the mobile economy. COVID funded the AI era. The rounds got harder but they never stopped. Click any crisis to see who got funded in the worst of times — and what they became.

Crises



- 9/11 attacks
- Asian financial crisis
- Dot-com crash
- European debt crisis
- Financial crisis
- Rate hike cycle begins
- Russian default and LTCM
- Taper tantrum

First Round VC Funding by Year and Sector.



Company	GICS Sector	Funding	Status	Round
Zyrra	Consumer Discretionary	\$6,042,000	operating	16
ZYOMYX	Health Care	\$137,100,060	operating	16
Zyngenia	Health Care	\$100,000,000	operating	4
Zynga	Information Technology	\$2,599,652,358	operating	27

\$1.44T
Total Funding (USD)

*VC data only available up to 2014

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Shifting Ground — And That Creates Opportunity



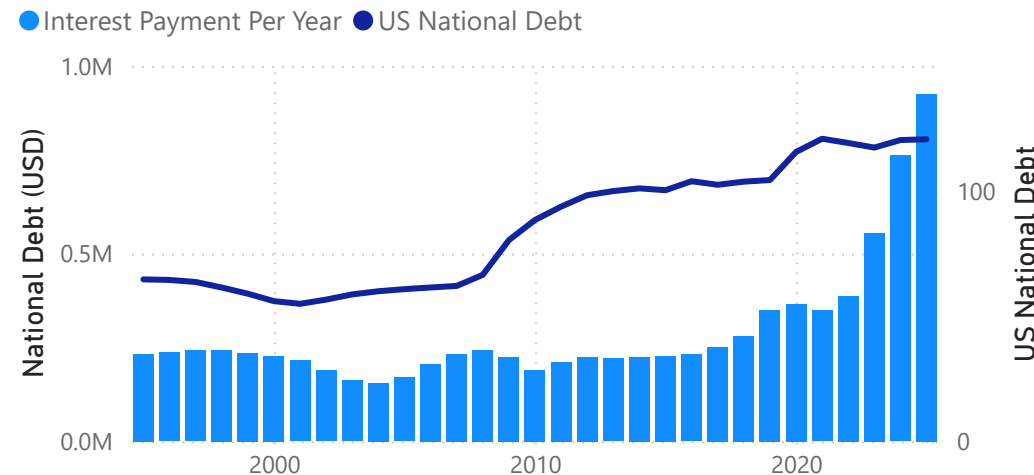
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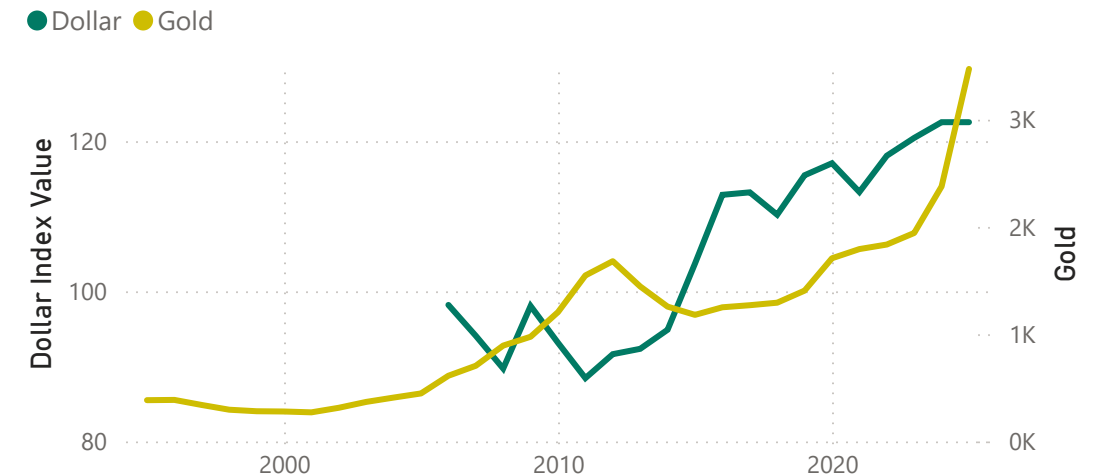
When debt becomes expensive, capital doesn't disappear — it rotates. Hard assets rise, sector leadership shifts, and the companies that move first capture the most value. As federal interest costs approach historic highs, gold's surge alongside the dollar index confirms the rotation is already underway — and history shows the winners are those who invest aggressively in what comes next.

The Debt Burden Is Becoming Unaffordable



US debt now exceeds 120% of GDP while annual interest payments approach \$1 trillion. The debt existed for decades but rising rates have made it exponentially more expensive to carry.

Hard Assets Are Replacing Paper Trust



Since 2020 gold has surged to all time highs while capital has rotated toward hard assets — a divergence that historically signals sophisticated investors repositioning ahead of structural market shifts.

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When The Rules Change, So Does The Playbook

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When the rules of the market change, the playbook changes with them. A decade of zero interest rates rewarded leverage, speculation, and duration. A world of structurally higher rates rewards cash flow, necessity, and hard assets. This framework isn't a prediction — it's a pattern. The investors who have always won are those who read the environment clearly and positioned accordingly. The opportunity in every crisis is the same: own what the next era needs before everyone else figures out what that is.

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SHORT

Sectors that depend on cheap credit and dollar stability:

Leveraged Consumer Discretionary — Depends on cheap credit and consumer confidence — both of which erode when rates stay high and purchasing power declines.

Commercial Real Estate — Built on cheap debt and remote work reversed the demand story — vacancy rates remain elevated while refinancing costs have doubled.

Late Stage Unprofitable Tech — Valuations were justified by zero interest rates — in a 4-5% rate environment the math simply doesn't work without a path to profitability.

Long Duration Bonds — When the creditworthiness of the issuer is being questioned and inflation remains sticky, holding long-dated government debt is the wrong side of the trade.

Long

Sectors that consolidate funding during stress and hold value in a new market environment:

Information Technology — Consolidates funding during every downturn in the dataset and benefits from AI-driven productivity gains that don't require cheap money to justify.

Health Care — Counter-cyclical demand that rises during economic stress and benefits from demographic tailwinds as the population ages.

Energy / Real Assets — Hard assets appreciate when the dollar weakens and energy remains essential regardless of economic conditions.

Gold — The market's oldest hedge against currency debasement — hitting all time highs precisely because institutional investors are losing confidence in paper assets.

**This represents an analytical framework for discussion purposes only and does not constitute investment advice.*